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Light Emitting Display Device

Abstract

Disclosed is a light emitting display device which may prevent or at least reduce lateral leakage current. The light emitting display device includes a substrate comprising a plurality of first to third subpixels arranged to neighbor one another, each of the first to third subpixels comprising an emission part and a non-emission part to surround the emission part; a bank at the non-emission parts; first electrodes covered with the bank and respectively at the first to third subpixels to expose the respective emission parts; a first spacer structure having an inversed tapered shape configured to surround at least one side of each of the emission parts of the first to third subpixels on the bank; a common layer structure located on the first electrodes and the first spacer structure and disconnected at an edge of the first spacer structure; and a second electrode on the common layer structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/345,864 filed on Jun. 11, 2021, which claims the benefit of Republic of Korea Patent Application No. 10-2020-0078445, filed on Jun. 26, 2020, each of which is hereby incorporated by reference in its entirety.

FIELD OF TECHNOLOGY

[0002] The present disclosure relates to a light emitting display device, and more particularly, to a light emitting display device which may reduce lateral leakage current between adjacent subpixels using a structure provided on a bank.

DISCUSSION OF THE RELATED ART

[0003] Recently, display devices have been developed into various types so as to satisfy requirements, such as flexibilization, miniaturization, maximization, etc.

[0004] Among display devices, a light emitting display device, which includes light emitting devices respectively provided in a plurality of pixels or subpixels provided on a substrate, and thus omits an external light source so as to realize slimness, is considered as a competitive application.

[0005] The light emitting display device includes a plurality of layers in each of the light emitting devices, and in this case, has common layers which are formed in common in the respective pixels or subpixels in order to avoid use of a fine metal mask.

[0006] Due to the common layers provided in common in the pixels or subpixels, current may flow to adjacent pixels or subpixels through the common layers extending two-dimensionally. Such current is referred to as lateral leakage current.

SUMMARY

[0007] Accordingly, the present disclosure is directed to a light emitting display device which may reduce lateral leakage current using a structure provided on a bank.

[0008] An object of the present disclosure is to provide a light emitting display device which may define a structure for separating common layers into regions corresponding to subpixels using a structure provided on a bank without a separate mask, and may thus disclosure lateral leakage current between adjacent subpixels.

[0009] Additional advantages, objects, and features of the invention will be set forth in part in the description which follows and in part will become apparent to those having ordinary skill in the art upon examination of the following or may be learned from practice of the invention. The objectives and other advantages of the invention may be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.

[0010] To achieve these objects and other advantages and in accordance with the purpose of the disclosure, as embodied and broadly described herein, a light emitting display device including a substrate including a plurality of first to third subpixels arranged to neighbor one another, each of

the first to third subpixels including an emission part and a non-emission part configured to surround the emission part, a bank provided in the non-emission parts, first electrodes covered with the bank and respectively provided in the first to third subpixels so as to expose the respective emission parts, a first spacer structure having an inverse tapered shape configured to surround at least one side of each of the emission parts of the first to third subpixels on the bank, a common layer structure located on the first electrodes and the first spacer structure and disconnected at an edge of the first spacer structure, and a second electrode located on the common layer structure.

[0011] In another aspect of the present disclosure, a light emitting display device includes a substrate including a plurality of emission parts and non-emission parts between the emission parts, first electrodes respectively provided in the emission parts, a bank configured to selectively overlap the first electrodes and provided in the non-emission parts, a first spacer structure having an inverse tapered shape configured to have open regions formed in a virtual closed loop configured to surround at least one of the emission parts and to surround at least one side of each of the emission parts, between adjacent emission parts on the bank, a common layer structure located on the first electrodes and the first spacer structure and disconnected at an edge of the first spacer structure, and a second electrode located on the common layer structure.

[0012] It is to be understood that both the foregoing general description and the following detailed description of the present invention are exemplary and explanatory and are intended to provide further explanation of the invention as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this application, illustrate embodiment(s) of the invention and together with the description serve to explain the principle of the invention. In the drawings:

[0014] FIG. 1 is a plan view illustrating a light emitting display device according to a first embodiment of the present disclosure;

[0015] FIG. 2 is a cross-sectional view taken along line I-I' of FIG. 1 according to the first embodiment of the present disclosure;

[0016] FIG. 3 is a cross-sectional view taken along line II-II' of FIG. 1 according to the first embodiment of the present disclosure;

[0017] FIG. 4 is a cross-sectional view taken along line III-III' of FIG. 1 according to the first embodiment of the present disclosure;

[0018] FIG. 5 is a plan view illustrating a light emitting display device according to a second embodiment of the present disclosure;

[0019] FIG. 6 is a plan view illustrating a light emitting display device according to a third embodiment of the present disclosure;

[0020] FIG. 7 is a plan view illustrating a light emitting display device according to a fourth embodiment of the present disclosure;

[0021] FIG. 8 is a plan view illustrating a light emitting display device according to a fifth embodiment of the present disclosure;

[0022] FIGS. 9A to 9E are plan views illustrating light emitting display devices according to sixth to tenth embodiments of the present disclosure, respectively;

[0023] FIGS. 10A to 10H are plan views illustrating light emitting display devices according to eleventh to eighteenth embodiments of the present disclosure, respectively;

[0024] FIG. 11 is a plan view illustrating a light emitting display device according to a nineteenth embodiment of the present disclosure;

[0025] FIGS. **12A** and **12B** are optical images of the light emitting display device according to the first embodiment of the present disclosure and a light emitting display device according to Example 1, when blue light is emitted at a low grayscale level; and

[0026] FIG. **13** is a scanning electron microscope (SEM) image representing a first spacer structure having an inverse tapered shape and elements therearound after formation of a first encapsulation layer, in the light emitting display device according to the preset disclosure.

DETAILED DESCRIPTION

[0027] Reference will now be made in detail to the exemplary embodiments of the present invention, examples of which are illustrated in the accompanying drawings. In the following description of the embodiments and the drawings, the same or similar elements are denoted by the same reference numerals throughout the specification. In the following description of the embodiments of the present invention, a detailed description of known functions and configurations incorporated herein will be omitted when it may make the subject matter of the present invention rather unclear. Further, the names of elements used in the following description of the embodiments of the present invention are selected in consideration of ease of preparation of the specification, and may thus differ from the names of parts of an actual product.

[0028] The shapes, sizes, ratios, angles and numbers of elements given in the drawings to describe the embodiments of the present invention are merely exemplary, and thus, the present invention is not limited to the illustrated details. In the following description of the embodiments, the terms “including”, “comprising” and “having” are to be interpreted as indicating the presence of one or more other characteristics, numbers, steps, operations, elements or parts stated in the specification or combinations thereof, and do not exclude the presence of other characteristics, numbers, steps, operations, elements, parts or combinations thereof, or the possibility of adding the same, unless the term “only” is used. It will be understood that a singular expression of an element(s) encompasses a plural expression unless stated otherwise.

[0029] In interpretation of elements included in the various embodiments of the present invention, it is to be interpreted that the elements include error ranges unless stated otherwise.

[0030] In the following description of the embodiments, it will be understood that, when positional relationships are expressed, for example, when an element is said to be “on”, “above”, “under” or “beside” another element, the two elements may directly contact each other, or one or more other elements may be interposed between the two elements, unless the term “just” or “directly” is used.

[0031] In the following description of the embodiments, it will be understood that, when the terms “first”, “second”, etc. are used to describe various elements, these terms are used merely to distinguish the same or similar elements. Therefore, a first element described hereinafter could be termed a second element without departing from the technical scope of the invention.

[0032] Respective features of the various embodiments of the present invention may be partially or wholly coupled to or combined with each other and be interlocked or driven in various technical manners, and the respective embodiments may be implemented independently of each other or be implemented together through connection therebetween.

[0033] FIG. **1** is a plan view illustrating a light emitting display device according to a first embodiment of the present disclosure. FIG. **2** is a cross-sectional view taken along line I-I' of FIG. **1**, FIG. **3** is a cross-sectional view taken along line II-II' of FIG. **1**, and FIG. **4** is a cross-sectional view taken along line III-III' of FIG. **1** according to the first embodiment of the present disclosure.

[0034] As shown in FIGS. **1** to **4**, the light emitting display device according to the first embodiment of the present disclosure includes a substrate **100** including a plurality of first to third subpixels SP**1**, SP**2** and SP**3** arranged to neighbor one another, each of the first to third subpixels SP**1**, SP**2** and SP**3** including an emission part B, R or G, and a non-emission part NE configured to surround the emission part B, R or G, a bank **150** provided in the non-emission parts NE, first electrodes **110a**, **110b** and **110c** covered with the bank **150** and respectively provided in the first to third subpixels SP**1**, SP**2** and SP**3** so as to expose the respective emission parts B, R and G, and a

first spacer structure **160a**, **160b** and **160c** having an inverse tapered shape configured to surround at least one side of a corresponding one of the emission parts B, R and G of the first to third subpixels SP1, SP2 and SP3 on the bank **150**.

[0035] Further, in the light emitting display device according to the present disclosure, light emitting devices, each of which includes the first electrode **110a**, **110b** or **110c**, a second electrode **180** provided opposite the first electrode **110a**, **110b** or **110c**, and an organic stack **121**, **122a** **122b** and **123** filling a space between the first electrode **110a**, **110b** or **110c** and the second electrode **180**, are provided in the respective subpixels SP1, SP2 and SP3. The light emitting device including a light emitting layer **122a** or **122b** formed of an organic light emitting material may be referred to as an organic light emitting device, the light emitting device including the light emitting layer **122a** or **122b** formed of a quantum dot material may be referred to as a quantum dot light emitting device, and the light emitting device including the light emitting layer **122a** or **122b** formed of a mixture of organic and inorganic materials may be referred to as a hybrid light emitting device. However, the light emitting devices of the light emitting display device according to the present disclosure are not limited in materials thereof. The light emitting display device according to the present disclosure includes one or more common layers **121** and **123** of a common layer structure which may be formed without a fine metal mask, and may be applied to self-light emitting devices having various structures as long as the common layers **121** and **123** may be separated into regions by the first spacer structure **160**: **160a**, **160b** and **160c** so as to prevent or reduce lateral leakage current between adjacent subpixels.

[0036] The first and second common layers **121** and **123** may be formed along the upper surface of the first spacer structure **160**: **160a**, **160b** and **160c**, and the upper surfaces of the first electrodes **110a**, **110b** and **110c** and the upper and side surfaces of the bank **150**, which are exposed outside the first spacer structure **160**: **160a**, **160b** and **160c**.

[0037] Further, the second electrode **180** may be formed on the second common layer **123**. The second electrode **180** including a metal material has relatively good step coverage, and may thus be formed not only on the upper part of the second common layer **123** but also on the side part of the second common layer **123**.

[0038] FIG. 1 illustrates one arrangement of the subpixels SP1, SP2 and SP3 of the light emitting display device according to the first embodiment of the present disclosure, and in this case, the first subpixels SP1 and the second subpixels SP2 may be alternately arranged in rows and columns, and the third subpixels SP3 may be arranged to alternate with each of the first subpixels SP1 and the second subpixels SP2 in a diagonal direction. In more detail, referring to FIG. 1, the first subpixels SP1 and the third subpixels SP3 are alternately arranged on a first diagonal line from upper left to lower right, and the second subpixels SP2 and the third subpixels SP3 are alternately arranged on a second diagonal line parallel to the first diagonal line. A set of one first subpixel SP1, one second subpixel SP2 and two third subpixels SP3 in a diamond region forms one pixel.

[0039] The first spacer structure **160** corresponding to at least one of the first to third subpixels SP1, SP2 and SP3 may have open regions **1600A**, **1600B** or **1600C** formed in a virtual closed loop configured to surround the emission part of the at least of the first to third subpixels SP1, SP2 and SP3. Although FIG. 1 illustrates the first spacer structure **160** corresponding to all of the first to third subpixels SP1, SP2 and SP3 as having the open regions **1600A**, **1600B** or **1600C** adjacent to the corresponding emission parts B, R and G, the disclosure is not limited thereto, and the first spacer structure **160** corresponding to one of the first to third subpixels SP1, SP2 and SP3 may have open regions, or the first spacer structure **160** corresponding to two of the first to third subpixels SP1, SP2 and SP3 may have open regions.

[0040] The present disclosure is characterized in that the first spacer structure **160**: **160a**, **160b** and **160c** having the inverse tapered shape is located on the bank **150** between adjacent subpixels SP1, SP2 and SP3 so that at least one of the common layers **121** and **123** deposited after formation of the first spacer structure **160**: **160a**, **160b** and **160c** may be disconnected at the side walls of the first

spacer structure **160**: **160a**, **160b** and **160c** having the inverse tapered shape. Thereby, even though the common layer has high conductivity, leakage current flowing to adjacent subpixels through the common layer may be prevented or at least reduced. Particularly, due to provision of the first spacer structure **160**: **160a**, **160b** and **160c**, a separate deposition mask configured to separate the common layers **121** and **123** into regions corresponding to the subpixels SP1, SP2 and SP3 may not be used in formation of the common layers **121** and **123**. Because the width of the upper side of the first spacer structure **160** is greater than the width of the lower side of the first spacer structure **160**, as seen in the cross-section thereof, organic matter forming the common layers **121** and **123** is not accumulated on a region of the bank **150** under the upper side of the first spacer structure **160**, and thus, the common layers **121** and **123** may be structurally separated into a region corresponding to the first spacer structure **160** and a region corresponding to the surroundings thereof as the boundary. That is, even though a space corresponding to a difference between the length of the upper side and the length of the lower side of the first spacer structure **160** is exposed from the bank **150**, the common layers **121** and **123** are not formed in this space because the upper side of the first spacer structure **160** functions as a screen. The common layer structure including common layers **121** and **123** is formed by supplying vaporized organic materials to the upper surface of the substrate **100** from a supply source provided above the substrate **100**. Because the vaporized organic materials having straightness are deposited, the common layers **121** and **123** are effectively formed on the upper surface of the first spacer structure **160** or on the upper surfaces of the first electrodes **110a** or **110b** or the upper surface of the bank **150** located outside the first spacer structure **160**, but are difficult to uniformly and continuously form on the regions of the bank **150** covered by the upper surface of the first spacer structure **160**, and thus the common layers **121** and **123** are structurally separated by the first spacer structure **160**.

[0041] As shown in FIG. 1, in the light emitting display device according to the first embodiment of the present disclosure, the first spacer structure **160** include first patterns **160b** configured to surround the emission parts B of the first subpixels SP1, second patterns **160a** configured to surround the emission parts R of the second subpixels SP2, and third patterns **160c** configured to surround the emission parts G of the third subpixels SP3. The first to third patterns **160b**, **160a** and **160c** may have the open regions **1600A**, **1600B** and **1600C** in the virtual closed loop shapes configured to surround the emission parts B, R and G of the first to third subpixels SP1, SP2 and SP3.

[0042] The first pattern **160b** provided around the emission part B of the first subpixel SP1 exhibits the following effects. In order to drive the first subpixel SP1, the first subpixel SP1 is turned on by applying a voltage equal to or higher than the threshold voltage V_{th} of the first subpixel SP1 thereto, and in this case, as shown in FIGS. 2 to 4, the common layers **121** and **123** are disconnected at the edge of the first pattern **160b**, and thus, little or no current leaks horizontally from the first subpixel SP1 towards adjacent subpixels and the second and third subpixels SP2 and SP3 having low threshold voltages are not turned on by leakage current. The second pattern **160a** structurally separates the first and second common layers **121** and **123**, and may thus prevent the second subpixel SP2 having the lower threshold voltage than the first and third subpixels SP1 and SP3 from being influenced by turning-on of adjacent first and third subpixels SP1 and SP3. The third pattern **160c** structurally separates the first and second common layers **121** and **123**, and may thus prevent the third subpixel SP3 having the lower threshold voltage than the first subpixel SP1 from being influenced by turning-on of an adjacent first subpixel SP1.

[0043] The threshold voltages of the first to third subpixels SP1, SP2 and SP3 are the minimum voltages required to drive thin film transistors (not shown) connected to the respective first electrodes **110a**, **110b** and **110c**. Although FIGS. 2 to 4 do not illustrate the thin film transistors, the thin film transistors, each of which includes a semiconductor layer, a gate electrode, a source electrode and a drain electrode, may be provided between the substrate **100** and the first electrodes **110a**, **110b** and **110c**.

[0044] The open regions **1600A**, **1600B** and **1600C** are provided in the first to third patterns **160b**, **160a** and **160c** so as to prevent or at least reduce an increase in linear resistance in a specific direction of the second electrode **180**. The second electrode **180** is provided in common in the subpixels **SP1**, **SP2** and **SP3**, and is formed without using a mask or is formed using an open mask configured to cover only the non-active area outside the subpixels **SP1**, **SP2** and **SP3**. The second electrode **180** is formed on the second common layer **123**, includes a metal so as to have good step coverage compared to the organic matter forming the common layers **121** and **123**, and may thus be formed not only on the upper part of the second common layer **123** but also on the side parts of the first and second common layers **121** and **123**. However, as the angle of the side part of the first spacer structure **160** from the surface of the bank **150** is decreased, a region, in which the second electrode **180** is not formed on the side part of the first spacer structure **160**, may occur. The light emitting display device according to the present invention may prevent an increase in resistance in a specific line direction of the second electrode **180** through the open regions **1600A** and **1600B** in the horizontal direction and the open region **1600C** in the vertical direction, even though the second electrode **180** is not formed on a portion of the side part of the first spacer structure **160**.

[0045] The width of the first spacer structure **160** is less than the critical dimension CD of the emission part having the smallest size among the emission parts B, R and G of the subpixels **SP1**, **SP2** and **SP3**. Further, the width of the first spacer structure **160** is narrow so that, even though a plurality of patterns are arranged between adjacent emission parts B, R and G, the patterns may be spaced apart from one another, and the length of the first spacer structure **160** may be greater than the length of at least one side of each of the emission parts B, R and G. The respective patterns **160b**, **160a** and **160c** forming the first spacer structure **160** is separated from one another so as to be distinguished from one another, and are provided around the emission parts B, R and G such that the length thereof is greater than the width thereof so as to prevent or reduce lateral leakage current from adjacent subpixels **SP1**, **SP2** and **SP3**. In the first embodiment, the third pattern **160c** has extensions between adjacent open areas **1600A** and **1600B** of the first and second patterns **160b** and **160a**, thereby being capable of effectively preventing or at least reducing leakage current from the first subpixel **SP1** to the second subpixel **SP2** among the first and second subpixels **SP1** and **SP2** having the maximum threshold voltage difference. The extension is located between the emission parts G of adjacent third subpixels **SP3** in the vertical direction, and may thus connect the third patterns **160c** surrounding the respective emission parts G. In this case, when the first subpixel **SP1** emits light, the path of current flowing in the first subpixel **SP1** towards the adjacent second subpixel **SP2** in the same row is blocked, the path of the current flowing in the first subpixel **SP1** directly towards the adjacent second subpixel **SP2** in the same column is also blocked by the first pattern **160b** and the second pattern **160a**, and thus, the current flowing in the first subpixel **SP1** makes a detour through the adjacent second subpixel **SP2** along the side part of the third pattern **160c**, the path of the current is elongated, and thereby, turning-on of the second subpixels **SP2** due to lateral leakage current may be prevented.

[0046] Here, the open regions **1600c** of the third pattern **160c** surrounding the emission part G of one third subpixel **SP3** may be located on different horizontal lines.

[0047] Further, in the light emitting display device according to the present disclosure, the first spacer structure **160** having the inverse tapered shape provided on the bank **150** are limited in terms of the length and width thereof so as to prevent or at least reduce an increase in sheet resistance of a specific region of the second electrode **180** formed after formation of the common layers **121** and **123**. Therefore, the second electrode **180** may maintain uniform voltage throughout the respective regions thereof. That is, the first spacer structure **160** has the open regions **1600A**, **1600B** and **1600C** for some of the emission parts B, R and G, thereby being capable of preventing or at least reducing an increase in the resistance of the second electrode **180** in a specific direction.

[0048] In the arrangement shown in FIG. 1, the first subpixels **SP1** have the largest size, the second subpixels **SP2** have the smallest size, the emission parts B and R of the first and second subpixels

SP1 and SP2 have an octagonal shape, the emission parts G of the third subpixels SP3 have a rectangular shape which extends in a diagonal direction and has rounded corners. Further, FIG. 1 illustrates that the third subpixels SP3 are arranged at a density two times that of the first and second subpixels SP1 and SP2. This arrangement is only one example, and the light emitting display device according to the present disclosure proposes a method for more effectively solving lateral leakage current in an arrangement structure in which subpixels are provided in a non-stripe type. The following embodiments will propose different arrangement structures. As will be described below, even in these different arrangement structures of subpixels, a first spacer structure **160** having an inverse tapered shape may also be arranged adjacent to emission parts so as to prevent or reduce lateral leakage current between the emission parts of adjacent subpixels.

[0049] The “inverse tapered shape” in the description of the present disclosure is a shape in which the length of the upper side of the first spacer structure **160**: **160a**, **160b** and **160c** is greater than the length of the lower side of the first spacer structure **160**: **160a**, **160b** and **160c**, as seen in the cross-section thereof, and thus, the side part of the first spacer structure **160** forms an acute angle from the surface of the bank **150** provided outside the first spacer structure **160**. In contrast, a second spacer structure **170** having a positive tapered shape, which is the opposite of the inverse tapered shape is provided, as shown in FIG. 2, and the “positive tapered shape” is a shape in which the length of the upper side of the second spacer structure **170** is smaller than the length of the lower side of the second spacer structure **170**, as seen in the cross-section thereof, and thus, the side part of the second spacer structure **170** forms an obtuse angle from the surface of the bank **150** provided outside the second spacer structure **170**.

[0050] The second spacer structure **170** is formed in a shape having a greater width and a greater height than the first spacer structure **160**: **160a**, **160b** and **160c**. The second spacer structure **170** primarily comes into contact with a deposition mask (not shown), used when the organic matter is deposited on the substrate **100** using the deposition mask, so as to support the deposition mask, thus protecting lower elements, for example, the bank **150** and the first spacer structure **160**: **160a**, **160b** and **160c**. The second spacer structure **170** may be formed on the bank **150** so as to be spaced apart from the first spacer structure **160**: **160a**, **160b** and **160c**. Because the second spacer structure **170** must uniformly support the deposition mask throughout the entirety of the substrate **100**, the second spacer structure **170** may be arranged in each of the subpixels SP1, SP2 and SP3 so as to have a designated width (area).

[0051] The above-described bank **150**, a first spacer structure **160** and second spacer structure **170** may be formed of organic insulating materials, such as photoacryl, polyimide and/or polyamide, and the bank **150**, the first spacer structure **160** and the second spacer structure **170** may include the same organic insulating material or at least one thereof may include a different organic insulating material. Further, in order to form different heights and different tapered shapes, the bank **150**, the first spacer structure **160** and the second spacer structure **170** may be formed through different processes, or the bank **150** and the second spacer structure **170** may be primarily formed of the same organic material using a half-tone mask and subsequently the first spacer structure **160** may be formed of a different organic material in regions spaced apart from the second spacer structure **170**.

[0052] In the light emitting display device according to the first embodiment of the present disclosure, in order to control lateral leakage current, the first spacer structure **160**: **160a**, **160b** and **160c** is not limited to a uniaxial shape in plan view, but have a shape surrounding the emission parts B, R and G and include the open regions **1600A**, **1600B** and **1600C** in the virtual closed loops configured to surround the emission parts B, R and G. Further, as shown in FIG. 1, the first subpixels SP1 and the second subpixels SP2 are alternately arranged in one row so that the open regions **1600A** and **1600B** of the first patterns **160b** and the second patterns **160a** are located on the same horizontal line, and the extensions of the third patterns **160c** pass between the first patterns **160b** and the second patterns **160a**, thereby being capable of preventing or reducing leakage

current on the horizontal line passing through the open regions **1600A** and **1600B** between the first and second subpixels **SP1** and **SP2** due to presence of the third patterns **160c** between the open regions **1600A** and **1600B**.

[0053] In some cases, the open regions **1600A** of the first patterns **160b** and the open regions **1600B** of the second patterns **160a** may be located on different horizontal lines so as to elongate the path of leakage current flowing on the horizontal lines.

[0054] The first electrodes **110a**, **110b** and **110c** may have protrusions protruding outwards from the respective emission parts **B**, **R** and **G**, and the protrusions may be connected to the corresponding thin film transistors (not shown).

[0055] The common layers **121** and **123** of the common structure are formed in common without distinction of the subpixels **SP1**, **SP2** and **SP3**. That is, these common layers **121** and **123** may be formed on the substrate **100** without using any mask, or may be formed on the substrate **100** using a common mask which covers the non-active area outside the subpixels **SP1**, **SP2** and **SP3** while opening all of the subpixels **SP1**, **SP2** and **SP3** on the substrate **100**. Therefore, use of fine metal masks, which have openings finely adjusted so as to correspond to emission parts of respective subpixels or specific subpixels, is reduced and thus an equipment burden may be remarkably reduced, and it is not necessary to align respective common layers with each other and thus yield may be remarkably improved.

[0056] The common layer structure including the common layers **121** and **123** may include, for example, an organic stack used to form the light emitting devices. The organic stack may include at least one of a hole injection layer, a hole transport layer, an electron blocking layer, an emission layer, a hole blocking layer, an electron transport layer, an electron injection layer and a charge generation layer. The light emitting device includes the first electrodes **110a**, **110b** or **110c**, the second electrode **180**, and the organic stack located therebetween.

[0057] The example shown in FIGS. 2 to 4 shows the first electrodes **110a** and **110b**, the second electrode **180**, and the first common layer **121**, emission layers **122a** and **122b** and the second common layer **123**, which are sequentially deposited on the substrate **100** in the upward direction, in the first and second subpixels **SP1** and **SP2**. Although not shown in these figures, the third subpixels **SP3** may have the same layer structure as the first and second subpixels **SP1** and **SP2**.

[0058] Although the illustrated example shows that the first common layer **121** is provided under the emission layers **122a** and **122b** as a single layer and the second common layer **123** is provided on the emission layers **122a** and **122b** as a single layer, the disclosure is not limited thereto, and the first common layer **121** may include different layers, such as a hole injection layer, a hole transport layer and an electron blocking layer, and the second common layer **123** may include different layers, such as a hole blocking layer and electron transport layer. Further, although the illustrated example shows that the emission layers **122a** and **122b** are provided as a single layer in the first and second subpixels **SP1** and **SP2**, a plurality of stacks divided from each other by a charge generation layer may be provided between the first electrodes **110a**, **110b** and **110c** and the second electrode **180**. Here, each “stack” means a configuration which essentially includes a hole transport layer, an emission layer and an electron transport layer.

[0059] The example of FIGS. 2 and 4 shows that the first emission layer **122a** configured to emit light with a first color is provided in each first subpixel **SP1** and the second emission layer **122b** configured to emit light with a second color different from the first color is provided in each second subpixel **SP2**, independently. The first and second emission layers **122a** and **122b** may be formed using a fine metal mask having openings corresponding to the emission parts **B** of the first subpixels **SP1** and the emission parts **R** of the second subpixels **SP2** so as to be selectively formed in the emission parts **B** of the first subpixels **SP1** and the emission parts **R** of the second subpixels **SP2**. Although not shown in these figures, an emission layer configured to emit light with a color different from the colors of light emitted by the first and second emission layers **122a** and **122b** may be provided in each third subpixel **SP3** in the same manner.

[0060] Although the example shown in FIGS. 2 to 4 represents the emission layers **122a** and **122b** as being independently provided in the respective subpixels so as to express colors of light emitted from the respective subpixels, the light emitting display device according to the present invention is not limited to thereto, and an emission layer may also be formed in common in a plurality of subpixels. In this case, in order to express various colors of light, a color filter layer or color conversion members may be further provided at the exit side of the light emitting display device separately from the emission layer.

[0061] The first subpixel SP1 may include the first electrode **110a**, the second electrode **180**, and the first emission layer **122a** configured to contact the first common layer **121** and the second common layer **123** between the first electrode **110a** and the second electrode **180** so as to emit light with a first color, the second subpixel SP2 may include the first electrode **110b**, the second electrode **180**, and the second emission layer **122b** configured to contact the first common layer **121** and the second common layer **123** between the first electrode **110b** and the second electrode **180** so as to emit light with a second color having a longer wavelength than the wavelength of the first color, and the third subpixel SP3 may include the first electrode **110c**, the second electrode **180**, and a third emission layer configured to contact the first common layer **121** and the second common layer **123** between the first electrode **110b** and the second electrode **180** so as to emit light with a third color having a wavelength between the wavelength of the first color and the wavelength of the second color.

[0062] Although the illustrated example of the light emitting display device according to the present disclosure represents that the first color is blue, the second color is red and the third color green, the disclosure is not limited to this color combination, and other color combinations may be implemented. In addition to the first to third colors, subpixels configured to emit white light or light with other colors rather than the first to third colors may be further provided.

[0063] Organic luminescent materials for respective colors require different threshold voltages, and among the blue, green and red organic luminescent materials which are currently known, the threshold voltage of the blue organic luminescent materials is highest, and the threshold voltage of the red organic luminescent materials is lowest. Therefore, in a light emitting display device having no structure for preventing lateral leakage current, such as the first spacer structure **160**, even when only blue subpixels are selectively turned on, other subpixels driven at threshold voltages lower than that of the blue subpixels may be turned on, and in this case, among red and green subpixels, the red subpixels driven at the lower threshold voltage may be more influenced by the turning-on of the blue subpixels. Depending on the threshold voltages of the colors of light emitted by the respective subpixels, in the light emitting display device according to the present invention, the first spacer structure **160** may be formed such that weight is assigned between the blue subpixel and the red subpixel, among the red and green subpixels adjacent to the blue subpixel.

[0064] FIG. 1 illustrates that, among the subpixels SP1, SP2 and SP3 shown in FIG. 1, the first subpixels SP1 have the largest size (e.g., area) and the second subpixels SP2 have the smallest size (e.g., area). The color of light emitted by the largest subpixels or the smallest subpixels may vary according to an application to which the light emitting display device is applied.

[0065] In the light emitting display device according to the present disclosure, the common layer structure including the first and second common layers **121** and **123** is structurally separated into regions on the first spacer structure **160** and regions on the surroundings thereof by the first spacer structure **160** without a fine metal mask, as described above, and, because the organic materials of the common layers **121** and **123** having high straightness are deposited, the first and second common layers **121** and **123** are not formed on the inverse tapered side surface of the first spacer structure **160**, and are thus be separated into regions corresponding to the adjacent subpixels.

[0066] The second electrode **180** formed after formation of the second common layer **123** is formed of a transparent or transfective component or an opaque metal component. The second electrode **180** having metal component has good step coverage compared to the first and second

common layers **121** and **123** formed of organic matter. The second electrode **180** may be formed on a portion of the side part of the first spacer structure **160**, and may be formed to cover not only the upper parts of the first and second common layers **121** and **123** but also the side parts of the first and second common layers **121** and **123** formed on the bank **150**.

[0067] In the light emitting display device according to the present disclosure, because the common layer structure including the first and second common layers **121** and **123** must be separated by the first spacer structure **160**, the height of the first spacer structure **160** may be greater than the total thickness of the organic stack formed between the first electrodes **110a**, **110b** or **110c** and the second electrode **180**. The organic stack includes the emission layer, and when the thicknesses of the organic stacks of the respective subpixels SP1, SP2 and SP3 are different, the first spacer structure **160** is formed to have a greater height than the thickness of the thickest organic stack. In order to prevent or at least reduce lateral leakage current due to the first spacer structure **160**, the width of the lower side of the first spacer structure **160** shown in FIGS. 2 to 4 may be 1 μm to 5 μm , and the height of the first spacer structure **160** may be equal to or greater than 1 μm which is less than the height of the second spacer structure **170**. The width of the upper side of the second spacer structure **170** may be 3 times to 10 times the width of the lower side of the first spacer structure **160**, and the height of the second spacer structure **170** may be equal to or less than 5 μm .

[0068] Although not shown in the drawings, a capping layer (not shown) may be further provided on the second electrode **180** in order to improve luminous efficacy, and an encapsulation layer formed by alternately at least one pair of an inorganic film and an organic film or an encapsulation substrate may be further provided on the capping layer in order to protect the light emitting devices from ambient air and to block moisture. When the encapsulation layer is applied, the encapsulation layer enters the side parts of the first and second spacer structure **160** and **170** on the bank **150**, and at least the organic film of the encapsulation layer has a thickness equal to or greater than the sum of the thickness of the bank **150** and the height of the second spacer structure **170**, and thus, the upper surface of the encapsulation layer planarizes the whole area of the light emitting display device including the subpixels.

[0069] FIG. 5 is a plan view illustrating a light emitting display device according to a second embodiment of the present disclosure.

[0070] As shown in FIG. 5, in the light emitting display device according to the second embodiment of the present disclosure, third patterns **160d** of a first spacer structure **160A** are arranged at a low density, and thus, third open regions **1600D** of the third patterns **160d** are relatively expanded, compared to the first embodiment. In this case, the density of the first spacer structure **160A** in horizontal rows, in which the third subpixels SP3 are arranged so as to neighbor one another, with respect to a second electrode **180** is decreased, compared to the first embodiment, and thus resistance of the second electrode **180** may be reduced, and first and third patterns **160b**, **160a** and **160d** are arranged so as to correspond to emission parts B, R and G and thus the effect in which, when a specific subpixel is turned on, the flow of current from the corresponding subpixel to the emission parts of adjacent subpixels is blocked may be exhibited, in the same manner as the first embodiment.

[0071] FIG. 6 is a plan view illustrating a light emitting display device according to a third embodiment of the present disclosure.

[0072] As shown in FIG. 6, in the light emitting display device according to the third embodiment of the present disclosure, in a first spacer structure **160B**, a second pattern **160e** configured to surround the emission part R of a second subpixel SP2 is divided into four pieces and thus has second open regions **1600B**, which are symmetrical to each other on a horizontal line, and first sub-open regions **1600E**, which are symmetrical to each other on a vertical line. Third patterns **160d** are arranged at a low density, in the same manner as those in the second embodiment, and thus resistance of a second electrode **180** in the horizontal direction may be reduced.

[0073] Further, the first sub-open regions **1600E** have a relatively large size, and first patterns **160b**

are arranged opposite the first sub-open regions **1600E**. Therefore, in a first subpixel **SP1** and a second subpixel **SP2** which are arranged vertically adjacent to each other, leakage current flowing from the first subpixel **SP1** directly to the second subpixel **SP2** in the vertical direction may be blocked by the first pattern **160b**, and leakage current flowing from the emission part B of the first subpixel **SP1** through a first open region **1600A** may make a detour through a space between the first pattern **160b** and the adjacent third pattern **160d**. Therefore, leakage current in the vertical direction may be prevented or reduced.

[0074] The effect in which leakage current flowing in the horizontal direction is blocked by the third pattern **160d** provided between the second open region **1600B** and the first open region **1600A** is the same as in the first embodiment.

[0075] That is, even in this case, the first to third patterns **160b**, **160e** and **160d** are configured to surround the corresponding emission parts B, R and G, and thus, the effect in which, when a specific subpixel is turned on, the flow of current from the corresponding subpixel to the emission parts of adjacent subpixels is blocked may be exhibited, in the same manner as the first embodiment.

[0076] Further, in the light emitting display device according to the third embodiment of the present invention, as shown in FIG. 6, due to additional provision of the first sub-open regions **1600E**, a space between the first and second subpixels **SP1** and **SP2** adjacent to each other in the vertical direction on the bank **150** is increased, and thus a margin of a region, in which a second spacer structure **170** will be arranged between the first and second subpixels **SP1** and **SP2** adjacent to each other in the vertical direction, may be secured.

[0077] FIG. 7 is a plan view illustrating a light emitting display device according to a fourth embodiment of the present disclosure.

[0078] As shown in FIG. 7, in the light emitting display device according to the fourth embodiment of the present disclosure, in a first spacer structure **160C**, a first pattern **160g** configured to surround the emission part B of a first subpixel **SP1** is divided into four pieces and thus has first open regions **1600A**, which are symmetrical to each other in the horizontal direction, and second sub-open regions **1600G**, which are symmetrical to each other in the vertical direction, in the same manner as second patterns **160e**, and a fourth open region **1600F** is further formed in the extensions of third patterns **160f** configured to surround the emission parts G of adjacent third subpixels **SP3**, compared to the third embodiment.

[0079] Because a distance between the emission parts B and R of the first and second subpixels **SP1** and **SP2** adjacent to each other in the vertical direction is long, a flow path of leakage current is long and thus the influence of leakage current may be reduced. Because the first open region **1600A** of the first pattern **160g**, the fourth open region **1600F** of the third pattern **160f**, and the second open region **1600B** of the second pattern **160e** are continuously located between the first subpixel **SP1** and the second subpixel **SP2** adjacent to each other in the horizontal direction, some leakage current in the horizontal direction may occur, compared to the first to third embodiments, but the fourth open region **1600F** is located at a position different from the positions of the first and second open regions **1600A** and **1600B**, thereby being capable of generating a bypass path of leakage current in the horizontal direction. Otherwise, a consecutive distance between the first open region **1600A**, the fourth open region **1600F** and the second open region **1600B**, which are continuously located, in the length direction of a horizontal direction or a vertical direction or a diagonal direction may be reduced so as to reduce the width of the path of the leakage current in the length direction.

[0080] Further, the light emitting display device according to the fourth embodiment of the present invention, as shown in FIG. 7, may increase a margin for arranging a second spacer structure **170** and reduce the resistance of a second electrode **180**, in the same manner as the third embodiment.

[0081] FIG. 8 is a plan view illustrating a light emitting display device according to a fifth embodiment of the present invention.

[0082] As shown in FIG. 8, in the light emitting display device according to the fifth embodiment of the present disclosure, a first spacer structure **160D** further includes bar-shaped fourth patterns **160h** between a first subpixel SP1 and a second subpixel SP2 adjacent to each other in the vertical direction, compared to the fourth embodiment. In this case, the fourth patterns **160h** may be provided in an island-type, and although this figure illustrates two fourth patterns **160h** provided between the first and second subpixels SP1 and SP2 adjacent to each other, one fourth pattern **160h** may be provided or three or more fourth patterns **160h** may be provided so as to increase the arrangement density.

[0083] The light emitting display device according to the fifth embodiment shown in FIG. 8 includes the fourth patterns **160h**, thus being capable of more effectively blocking leakage current in the vertical direction, compared to the fourth embodiment.

[0084] Hereinafter, other modified embodiments will be described.

[0085] FIGS. 9A to 9E are plan views illustrating light emitting display devices according to sixth to tenth embodiments of the present invention.

[0086] As shown in FIG. 9A, in the light emitting display device according to the sixth embodiment of the present disclosure, a first spacer structure **260A** includes first patterns **260b** configured to surround emission parts B of first subpixels SP1, second patterns **260a** configured to surround emission parts R of second subpixels SP2, and third patterns **260c** and **260d** configured to surround emission parts G of third subpixels SP3.

[0087] Here, the first patterns **260b** and the second patterns **260a** include first open regions **2600A** and second open regions **2600B** which are not located on the same horizontal line. Thereby, the effect of blocking leakage current in the horizontal direction may be increased.

[0088] Further, the third patterns **260c** and **260d** may be divided into first subpatterns **260c** adjacent to the first patterns **260b** and second subpatterns **260d** adjacent to the second patterns **260a**. The first subpatterns **260c** and the second subpatterns **260d** may have third open regions **2600C** between the first and second subpatterns **260c** and **260d** adjacent to the emission parts G in the vertical direction. In this case, two third open regions **2600C** located at the left and right sides of the emission part G may be provided on different horizontal lines. Further, the first and second subpatterns **260c** and **260d** have a shape which extends to be bent in the outward direction of the emission part G. Further, the first subpatterns **260c** and the second subpatterns **260d** may have fourth open regions **2600D** outside the emission parts G, and in this case, two third open regions **2600D** located at the left and right of the emission part G may also be provided on different horizontal lines.

[0089] The first open region **2600A**, the fourth open regions **2600D** and the second open region **2600B** are located on different horizontal lines between the first and second subpixels SP1 and SP2 adjacent to each other in the horizontal direction, so as to form a zigzag-shaped path of leakage current, thereby being capable of increasing the length of the path of leakage current and thus reducing generation of lateral leakage current when a specific subpixel is turned on.

[0090] As shown in FIG. 9B, in the light emitting display device according to the seventh embodiment of the present disclosure, a first spacer structure **260B** includes first to third patterns **260f**, **260e** and **260g** configured to surround respective emission parts B, R and G of first to third subpixels SP1, SP2 and SP3, the first patterns **260f** have fifth open regions **2600E**, the second patterns **260e** have sixth open regions **2600F**, and the third patterns **260g** have seventh open regions **2600G**.

[0091] In this case, when a distance among the emission parts B, R and G of the subpixels SP1, SP2 and SP3 is sufficiently long, the effect of blocking leakage current may be exhibited.

[0092] As shown in FIG. 9C, in the light emitting display device according to the eighth embodiment of the present disclosure, a first spacer structure **260C** is configured such that an eighth open region **2600H** is formed in only one side of a first pattern **260i** surrounding the emission part B of each of first subpixels SP1, and a ninth open region **2600I** is formed in only one

side of a second pattern **260h** surrounding the emission part R of each of second subpixels SP2. In this case, each of the first patterns **260i** and the second patterns **260h** is not divided, and is provided in an integral type. Further, third patterns **260j** configured to surround the emission parts G of third subpixels SP3 extend so as to be bent upwards and downwards from the emission parts G in the outward direction, and thus passes between the first pattern **260i** and the second pattern **260h** adjacent to each other. The third patterns **260j** are separately formed at both sides of the emission parts G, and may have tenth open regions **2600J** which are formed in extensions thereof outside the emission parts G so as to be located on different horizontal lines parallel to each other.

[0093] In the light emitting display device according to the eighth embodiment, the open regions **2600H**, **2600I** and **2600J** are not continuously provided in both the horizontal direction and the vertical direction, thereby being capable of effectively blocking leakage current.

[0094] As shown in FIG. **9D**, in the light emitting display device according to the ninth embodiment of the present disclosure, a first spacer structure **260D** further includes closed loop-shaped fifth patterns **260m**, which are spaced apart from first patterns **260i** and second patterns **260h**, between first and second subpixels SP1 and SP2 adjacent to each other in the vertical direction, compared to the eighth embodiment. In this case, because, at least between the first and second subpixels SP1 and SP2 adjacent to each other in the vertical direction, current flowing in the emission part B of the first subpixel SP1 flows out through an eighth open region **2600H** and passes around the fifth pattern **260m**, and flows into a ninth open region **2600I** formed at one side of the emission part R of the second subpixel SP2, the path of leakage current is elongated, compared to the eighth embodiment, and thus, there is little influence of leakage current between the first and second subpixels SP1 and SP2.

[0095] Further, between the first subpixel SP1 and the third subpixel SP3, current flowing in the emission part B of the first subpixel SP1 flows out through the eighth open region **2600H**, passes around the fifth pattern **260m**, and flows along the shape of the bent part of a tenth open region **2600J** of a third pattern **260k**, located opposite the eighth open region **2600H**, there is little influence of leakage current between the first and third subpixels SP1 and SP3 due to such current path.

[0096] As shown in FIG. **9E**, in the light emitting display device according to the tenth embodiment of the present disclosure, a first spacer structure **260E** further includes first protrusion patterns **260j** and second protrusion patterns **260n**, which protrude from the first patterns **260h** and the second patterns **260i**, configured to surround the emission parts B and R of the first and second subpixels SP1 and SP2, to each other so as to be spaced apart from each other, compared to the eighth embodiment. In this case, the first protrusion pattern **260j** is spaced apart from the adjacent second pattern **260i** by an eleventh open region **2600L**, and the second protrusion pattern **260n** is spaced apart from the adjacent first pattern **260h** by a twelfth open region **2600K**.

[0097] In the light emitting display device according to the tenth embodiment, current flowing outwards from the first subpixel SP1 through the eighth open region **2600H** passes between the first pattern **260h** and the third pattern **260k** adjacent to each other, passes between the first protrusion pattern **260j** and the third pattern **260k** adjacent to each other, passes through the eleventh open region **2600L**, is bent to pass between the first and second protrusion patterns **260j** and **260n**, passes through the twelfth pattern **2600K**, passes through the second protrusion pattern **260n** and the third pattern **260k** adjacent to each other, passes along the second pattern **260h**, and then passes through the ninth open region **2600I** of the second pattern **260h**. In this case, the flow path of current is excessively long, and thus influence of leakage current between the first and second subpixels SP1 and SP2 may be prevented.

[0098] Current flowing between the first subpixel SP1 and the third subpixel SP3 also passes through a zigzag-shaped path through the eleventh and twelfth open regions **2600L** and **2600K** between the first and second protrusion patterns **260j** and **260n** and the second pattern **260h** and the first pattern **260i** adjacent thereto, and thus, current flowing out from the first subpixel SP1 may

scarcely influence the third subpixel SP3 adjacent thereto.

[0099] Hereinafter, the shapes of first spacer structures in other embodiments having structures and arrangements of subpixels differing from the above-described structure and arrangement of the subpixels will be described.

[0100] FIGS. **10A** to **10H** are plan views illustrating light emitting display devices according to eleventh to eighteenth embodiments of the present disclosure.

[0101] In common in the light emitting display devices according to the eleventh to eighteenth embodiments, which will be described below, first subpixels SP1 are arranged in a first axis direction (in the vertical direction in these figures), second subpixels SP2 and third subpixels SP3 are alternately arranged in the first axis direction, and the first subpixel SP1 has a shape configured to neighbor at least one of the second subpixel SP2 and the third subpixel SP3 in a second axis direction (in the horizontal direction in these figures) intersecting the first axis direction.

[0102] Further, the first to third subpixels SP1, SP2 and SP3 respectively include emission parts B, R and G formed therein, and regions of the first to third subpixels SP1, SP2 and SP3, which are exposed from a bank **350**, may be referred to as the emission parts B, R and G.

[0103] In the first to third subpixels SP1, SP2 and SP3, first electrodes **310a**, **310b** and **310c**, which overlap the corresponding emission parts B, R and G and overlap the bank **350** at the edges thereof, are provided.

[0104] The first electrodes **310a**, **310b** and **310c** of the first to third subpixels SP1, SP2 and SP3 may be connected to thin film transistors provided on the substrate **400** (with reference to FIGS. **2** to **4**) through connection parts CT1, CT2 and CT3.

[0105] As shown in FIG. **10A**, in the light emitting display device according to the eleventh embodiment of the present disclosure, a first spacer structure **360A** may include first patterns **360a** arranged in the first axis direction (i.e., the vertical direction) between the first subpixels SP1 and the second subpixels SP2, and second patterns **360b** and **360c** arranged between the second subpixels SP2 and the third subpixels SP3. The second patterns **360b** and **360c** may be arranged in the second axis direction (i.e., the horizontal direction) intersecting the first axis direction.

[0106] The first patterns **360a** may have first open regions **3600A** adjacent to the emission parts B of the first subpixels SP1, and the first open regions **3600A** may be adjacent to the emission parts G of the third subpixels SP3.

[0107] Further, the second patterns **360b** and **360c** may have second open regions **3600B** between the second and third subpixels SP2 and SP3 adjacent to each other. The second patterns **360b** and **360c** may be divided into primary patterns **360b** which are adjacent to the centers of the emission parts B of the first subpixels SP1 and extend between the second and third subpixels SP2 and SP3, and secondary patterns **360c** which are adjacent to the edges of the emission parts B of the first subpixels SP1 and extend between the second and the third subpixels SP2 and SP3.

[0108] Further, the first patterns **360a** may have third open regions **3600C** between the first subpixels SP1 and the second subpixels SP2. It may be more effective in terms of blocking of leakage current flowing from the first subpixel SP1 to the second subpixel SP2 to locate the third open regions **3600C** in parts of the first patterns **360a** which are not adjacent to the emission parts R of the second subpixel SP2.

[0109] Here, the first patterns **360a** and the second patterns **360b** and **360c** may be connected to each other. The connection parts therebetween may be formed at the intersections between the first subpixels SP1 and the second and third subpixels SP2 and SP3. Therefore, the first spacer structure **360A** may be arranged at the intersections, and thus, when the first subpixel SP1 is turned on, the shortest paths of current from the first subpixel SP1 to the second and third subpixels SP2 and SP3 are blocked and thereby the effect of preventing or at least reducing leakage current may be improved.

[0110] The first and third open regions **3600A** and **3600C**, which are provided in the first patterns **360a**, and the second open regions **3600B**, which are provided in the second patterns **360b** and

360c, serve to prevent or reduce an increase in resistance of the second electrode **180** (with reference to FIGS. **2** to **4**), formed on the organic stacks **121**, **122a/122b** and **123** (with reference to FIGS. **2** to **4**), in the vertical direction.

[0111] The light emitting display device according to this embodiment diverts current flowing out from the center of the emission part B of the first subpixel SP1, and may thus mitigate the influence of turning-on of the first subpixel SP1 on driving of the second and third subpixels SP2 and SP3 adjacent to the first subpixel SP1.

[0112] As shown in FIG. **10B**, in the light emitting display device according to the twelfth embodiment of the present disclosure, a first spacer structure **360B** is configured such that second patterns **360d** have no open region so as to completely isolate the emission parts R and G of the second and third subpixels SP2 and SP3 from each other, compared to the eleventh embodiment.

[0113] In this case, the light emitting display device according to the twelfth embodiment is further advantageous in that leakage current from the emission part G of the third subpixel SP3 having relatively high threshold voltage to the emission part R of the second subpixel SP2 is blocked.

[0114] As shown in FIG. **10C**, in the light emitting display device according to the thirteenth embodiment of the present disclosure, a first spacer structure **360C** is configured such that first patterns **360f** have no open region so as to completely isolate the first subpixels SP1 and the second and third subpixels SP2 and SP3 from each other, compared to the eleventh embodiment.

[0115] In this case, the light emitting display device according to the thirteenth embodiment is further advantageous in that leakage current from the emission part B of the first subpixel SP1 having the highest threshold voltage to the emission parts R and G of the second and third subpixels SP2 and SP3 is blocked.

[0116] As shown in FIG. **10D**, in the light emitting display device according to the fourteenth embodiment of the present disclosure, a first spacer structure **360D** further includes first auxiliary patterns **360g** parallel to first patterns **360a**, and two second patterns **360c1** and **360c2** are connected to the first pattern **360a**, compared to the eleventh embodiment. The light emitting display device according to the fourteenth embodiment may have similar effects of preventing or reducing leakage current and reducing resistance of the second electrode to the eleventh embodiment.

[0117] As shown in FIG. **10E**, in the light emitting display device according to the fifth embodiment of the present disclosure, a first spacer structure **360E** further includes first auxiliary patterns **360h** parallel to first patterns **360a**, and two second patterns **360c1** and **360c2** are connected to the first pattern **360a**, compared to the twelfth embodiment. Compared to the fourteenth embodiment, no open regions are formed in the second patterns **360c1** and **360c2**, and thus, the effect of blocking leakage current between the second and third subpixels SP2 and SP3 may be improved.

[0118] Further, fourth and fifth open regions **3600D** and **3600E** formed in the first auxiliary patterns **360h** adjacent to the emission parts B of the first subpixels SP1 are located on horizontal lines different from the horizontal lines on which the first and third open regions **3600A** and **3600C** are formed, and thereby the direct path of current is blocked and thus the effect of preventing or reducing leakage current may be improved.

[0119] As shown in FIG. **10F**, in the light emitting display device according to the sixteenth embodiment of the present disclosure, a first spacer structure **360E** further includes first auxiliary patterns **360f** having no open regions parallel to first patterns **360a**, and two second patterns **360c1** and **360c2** are connected to the first pattern **360a**, compared to the fifth embodiment. Compared to the fifteenth embodiment, no open regions are formed in the first auxiliary patterns **360f**, and thus, leakage current making a detour from the emission part B of the first subpixel SP to the emission part R of the second subpixel SP2 and the emission part G of the third subpixel SP3 may be completely blocked.

[0120] As shown in FIG. **10G**, in the light emitting display device according to the seventeenth

embodiment of the present disclosure, a first spacer structure **360G** may include first patterns **360i** and first auxiliary patterns **360j** arranged in parallel in the first axis direction (the vertical direction) between the first subpixels **SP1** and the second subpixels **SP2**, and second patterns **360c1** and **360c2** arranged in parallel in the second axis direction intersecting the first axis direction between the second subpixels **SP2** and the third subpixels **SP3**. The second patterns **360c1** and **360c2** may have second open regions **3600B** adjacent to each other.

[0121] The first patterns **360i** and the first auxiliary patterns **360j** may respectively have first open regions **3600A** and fourth open regions **3600D** which are not adjacent to each other. In the light emitting display device according to the seventeenth embodiment, current flowing out from the emission part **B** of the first subpixel **SP1** makes a detour to the emission part **G** of the third subpixel **SP3** through the first and fourth open regions **3600D** and **3600A** which are not adjacent to each other, and thereby, an influence of leakage current from the first subpixel **SP1** on the third subpixel **SP3** may be minimized.

[0122] Further, a direct path of current from the first subpixel **SP1** to the second subpixel **SP2** in the horizontal and vertical directions is blocked, and thereby, an influence of leakage current from the first subpixel **SP1** on the second subpixel **SP2** may be completely blocked.

[0123] As shown in FIG. **10H**, in the light emitting display device according to the eighteenth embodiment of the present disclosure, a first spacer structure **360G** may include first patterns **360a**: **360a1** and **360a2** and first auxiliary patterns **360j** arranged in parallel in the first axis direction (the vertical direction) between the first subpixels **SP1** and the second subpixels **SP2**, and second patterns **360b1** and **360b2** arranged in parallel in the second axis direction intersecting the first axis direction between the second subpixels **SP2** and the third subpixels **SP3**. The second patterns **360c1** and **360c2** may have second open regions **3600B** adjacent to each other.

[0124] The first patterns **360a** are divided into first subpatterns **360a1** and second subpatterns **360a2**, and may thus have a first open region **3600A** adjacent to the emission part **G** of the third subpixel **SP3** between the first subpattern **360a1** and the second subpatterns **360a2**, and a third open region **3600C** adjacent to the emission part **R** of the second subpixel **SP2** between the second subpatterns **360a2** and the first subpattern **360a1**. Further, the first patterns **360a** may further include connection parts **360k** configured to connect the first patterns **360a1** and **360a2** and the first auxiliary patterns **360j**, which are parallel to each other. The first auxiliary patterns **360j** may have fifth open regions **3600E** arranged on a horizontal line different from the first and third open regions **3600A** and **3600C**. The connection parts **360k** are located between the first patterns **360a1** and **360a2** and the first auxiliary patterns **360j**, which are parallel to each other, thus being capable of blocking leakage current or guiding the flow of current.

[0125] In the light emitting display device according to the eighteenth embodiment shown in this figure, the path of current flowing out from the emission part **B** of the first subpixel **SP1** makes a detour to the second subpixel **SP2** through the fifth and third open regions **3600D** and **3600C** which are not adjacent to each other, and thereby, an influence of leakage current from the first subpixel **SP1** on the second subpixel **SP2** may be minimized.

[0126] Further, a direct path of current from the first subpixel **SP1** to the third subpixel **SP3** in the horizontal and vertical directions is blocked, and thereby, an influence of leakage current from the first subpixel **SP1** on the third subpixel **SP3** may be completely blocked.

[0127] In the light emitting display devices according to the eleventh to eighteenth embodiments, a second spacer structure **170** (with reference to FIG. **1**) having a greater area and a greater height than the first spacer structure **360A** to **360G** may be formed. The second spacer structure **170** is spaced apart from each of the first spacer structures **360A** to **360G** and thus primarily contacts with a deposition mask so as to support the deposition mask, thus protecting elements (for example, the bank and the first spacer structure) provided under the second spacer structure.

[0128] FIG. **11** is a plan view illustrating a light emitting display device according to a nineteenth embodiment of the present disclosure.

[0129] As shown in FIG. 11, in the light emitting display device according to the nineteenth embodiment of the present disclosure, a first spacer structure **60** is formed so as to completely surround emission parts B, R and G of respective subpixels SP1, SP2 and SP3. In this case, an influence of the turn-on operation of each subpixel on other subpixels may be prevented, but resistance of a second electrode (cathode), which must be formed on the entirety of the surfaces of the subpixels, in the horizontal and vertical direction may be increased due to horizontal and vertical connection of the first spacer structure **60**. Therefore, the light emitting display devices according to the eleventh to eighteenth embodiments of the present disclosure, in which open regions are provided on at least one of the emission parts, may be desirable from the point of view of maintenance of uniform voltage in the second electrode.

[0130] Hereinafter, Test Example 1 which is implemented so as to comparatively describe the effects exhibited by the first spacer structure of the light emitting display device according to the present invention will be described.

[0131] FIGS. 12A and 12B are optical images of a light emitting display device according to Test Example 1 such as the first embodiment of the present disclosure and the light emitting display device according to the Test Example 2, respectively, when blue light is emitted at a low grayscale level.

[0132] Hereinafter, the emission states of the light emitting display device having the first spacer structure **160** according to the Test Example 1 such as the first embodiment of the present disclosure and the light emitting display device having no first spacer structure according to Test Example 2, when only blue subpixels thereof were turned on, were observed through a microscope.

[0133] As shown in FIG. 12A, it may be confirmed that, in the light emitting display device according to the Test Example 1 such as the first embodiment, in which the first spacer structure **160** having an inverse tapered shape is provided, only the blue subpixels emit light. Further, it may be confirmed that, because the first spacer structure has the open regions, voltage is uniformly applied to the second electrode (cathode) and thus the blue subpixels effectively emitted light.

[0134] In contrast, as shown in FIG. 12B, it may be confirmed that, in the light emitting display device of Test Example 2, in which no first spacer structure having an inverse tapered shape is provided, when the blue subpixels **1201** were turned on, the red subpixels **1203** having low threshold voltage around the blue subpixels were turned on. That is, undesired subpixels were turned on due to lateral leakage current, thus causing deterioration in image quality.

[0135] FIG. 13 is an SEM image representing the first spacer structure having an inverse tapered shape and the elements therearound after formation of the first encapsulation layer, in the light emitting display device according to the present invention.

[0136] FIG. 13 shows the first spacer structure and the elements therearound of the light emitting display device according to the present disclosure, after formation of the first encapsulation layer, and it may be confirmed that the organic stack is not formed on the side part of the first spacer structure, but is formed on the flat upper part of the first spacer structure and on the upper surface of the bank, and thus, the organic stack is separated into respective regions by the upper part of the first spacer structure and the upper surface of the bank having a step therebetween. The first encapsulation layer, which is formed of an inorganic material on the organic stack, has a thickness equal to or greater than about 1 m, and it may be confirmed that the first encapsulation layer formed of an inorganic material is formed to have continuity, and is thus continuously formed not only on the upper part of the first spacer structure and the surface of the bank not only on the side part of the first spacer structure.

[0137] This means that the organic stack is separated into the regions corresponding to respective subpixels due to provision of the first spacer structure and, after formation of the second electrode, the first encapsulation layer is formed to cover the first spacer structure without disconnection, and thus, the light emitting display device is normally encapsulated.

[0138] After formation of the first encapsulation layer, the second encapsulation layer formed of an

organic material may be further provided on the light emitting display device so as to planarize all regions of the light emitting display device.

[0139] In the light emitting display device according to each of the above-described embodiments of the disclosure invention, the first spacer structure having an inverse tapered shape is located on the bank between adjacent subpixels, and the common layers deposited after formation of the first spacer structure are disconnected at the first spacer structure having the inverse tapered shape, thereby being capable of blocking lateral leakage current caused by the common layers having high conductivity.

[0140] Further, the first spacer structure having the inverse tapered shape provided on the bank is limited in terms of the length and width thereof, thereby being capable of preventing or at least reducing an increase in the sheet resistance of a specific region of the second electrode (cathode) formed after formation of the common layer structure. Therefore, the second electrode may maintain uniform voltage. That is, the first spacer structure has open regions with respect to at least some of the emission parts, thus being capable of preventing or at least reducing an increase in the resistance of the second electrode.

[0141] Moreover, as the pixel structures of display devices are diversified, the first spacer structure is not limited to a uniaxial shape but has a shape surrounding the emission parts and including open regions formed in some parts of closed loop-type patterns configured to surround the emission parts so as to additionally guide the flow of current, thereby increasing the distance between the open regions of the first spacer structure between adjacent emission parts and thus being capable improving the effect of preventing or at least reducing lateral leakage current.

[0142] Further, in addition to the first spacer structure, a second spacer structure having a positive tapered shape is provided on the bank so as to have a greater height than the first spacer structure, thereby being capable of supporting a deposition mask when the deposition mask is placed on the substrate so as to form light emitting devices, and thus protecting the elements provided under the second spacer structure.

[0143] Furthermore, the first spacer structure having the inverse tapered shape is disposed at least between adjacent subpixels configured to emit light with different colors and to have a threshold voltage difference therebetween, so that the common layers are disconnected at the outside of the first spacer structure, thereby preventing, when the subpixel having high threshold voltage is selectively turned on, the subpixel having low threshold voltage arranged adjacent to the turned-on subpixel from being turned on due to lateral leakage current.

[0144] For this purpose, a light emitting display device according to one embodiment the present disclosure may include a substrate including a plurality of first to third subpixels arranged to neighbor one another, each of the first to third subpixels including an emission part and a non-emission part configured to surround the emission part, a bank provided in the non-emission parts, first electrodes covered with the bank and respectively provided in the first to third subpixels so as to expose the respective emission parts, a first spacer structure having an inverse tapered shape configured to surround at least one side of each of the emission parts of the first to third subpixels on the bank, a common layer structure located on the first electrodes and the first spacer structure and disconnected at an edge of the first spacer structure, and a second electrode located on the common layer structure.

[0145] The first spacer structure corresponding to at least one of the first to third subpixels may have open regions formed in a virtual closed loop configured to surround the emission part of the at least of the first to third subpixels.

[0146] The light emitting display device may further include a second spacer structure having a positive tapered shape spaced apart from the first spacer structure and configured to have a greater height than the first spacer structure.

[0147] The first spacer structure may include first patterns configured to surround the emission parts of the first subpixels, second patterns configured to surround the emission parts of the second

subpixels, and third patterns configured to surround the emission parts of the third subpixels, and the first to third patterns may be spaced apart from one another.

[0148] The first patterns and the second patterns respectively may have open regions provided so as not to neighbor each other.

[0149] The first spacer structure may have open regions formed in the first patterns and open regions formed in the second patterns between the emission parts of the first subpixels and the emission parts of the second subpixels, and extensions connected to the third patterns to block the open regions formed in the first and second patterns may be provided between the open regions formed in the first and second patterns.

[0150] The first spacer structure may further include fourth patterns provided between the first patterns and the second patterns so as to be spaced apart from the first and second patterns.

[0151] The first spacer structure may further include, between the first patterns and the second patterns, first protrusions configured to protrude from the first patterns so as to be spaced apart from the second patterns by a first distance, and second protrusions configured to protrude from the second patterns so as to be spaced apart from the first patterns by a second distance.

[0152] The first spacer structure may further include connection parts configured to connect the third patterns between adjacent third subpixels.

[0153] At least one of the first to third patterns may have a plurality of open regions, and may be divided by the open regions serving as boundaries.

[0154] The first subpixels and the second subpixels may be alternately arranged in rows and columns, and the third subpixels may be arranged to alternate with each of the first subpixels and the second subpixels in a diagonal direction.

[0155] The first subpixels may be arranged in a first axis direction, the second subpixels and the third subpixels may be alternately arranged in the first axis direction, and the first subpixel may neighbor at least one of the second subpixel and the third subpixel in a second axis direction.

[0156] The first spacer structure may include first patterns arranged in the first axis direction between the first subpixels and the second subpixels, and second patterns arranged between the second subpixels and the third subpixels.

[0157] At least one of the first patterns and the second patterns may have an open region adjacent to one of the emission parts of the first to third subpixels.

[0158] At least one of the first patterns and the second patterns may include a plurality of subpatterns arranged parallel to each other.

[0159] The subpatterns arranged parallel to each other may have open regions provided so as not to neighbor each other.

[0160] The light emitting display device may further include guide parts configured to connect the subpatterns arranged parallel to each other.

[0161] The first patterns and the second patterns may be connected to each other, and contact points between the first patterns and the second patterns may neighbor the emission parts of the first subpixels.

[0162] The first subpixels may have a first emission layer configured to contact the common layer structure between the first and second electrodes so as to emit light with a first color, the second subpixels may have a second emission layer configured to contact the common layer structure between the first and second electrodes so as to emit light with a second color having a longer wavelength than a wavelength of the first color, and the third subpixels may have a third emission layer configured to contact the common layer structure between the first and second electrodes so as to emit light with a third color having a wavelength between the wavelength of the first color and a wavelength of the second color.

[0163] The first spacer structure may have a width less than a critical dimension of an emission part having the smallest size among the emission parts of the first to third subpixels.

[0164] The common layer structure may include at least one of a hole injection layer, a hole

transport layer, an electron blocking layer, an emission layer, a hole blocking layer, an electron transport layer, an electron injection layer and a charge generation layer.

[0165] Among the first to third subpixels, a threshold voltage of the first subpixel may be highest, and a threshold voltage of the second subpixel may be lowest.

[0166] Among the first to third subpixels, an area of the first subpixel may be largest, and an area of the second subpixel may be smallest.

[0167] A light emitting display device according to another embodiment the present invention may include a substrate including a plurality of emission parts and non-emission parts between the emission parts, first electrodes respectively provided in the emission parts, a bank configured to selectively overlap the first electrodes, and provided in the non-emission parts, a first spacer structure having an inverse tapered shape configured to have open regions formed in a virtual closed loop configured to surround at least one of the emission parts and to surround at least one side of each of the emission parts, between adjacent emission parts on the bank, a common layer structure located on the first electrodes and the first spacer structure and disconnected at an edge of the first spacer structure, and a second electrode located on the common layer structure.

[0168] As is apparent from the above description, a light emitting display device according to the present invention has the following effects.

[0169] First, in the light emitting display device according to the present disclosure, a first spacer structure having an inverse tapered shape is located on a bank between adjacent subpixels, and common layers deposited after formation of the first spacer structure are disconnected at the first spacer structure having the inverse tapered shape, thereby being capable of blocking lateral leakage current caused by the common layers having high conductivity.

[0170] Second, the first spacer structure having the inverse tapered shape provided on the bank is limited in terms of the length and width thereof, thereby being capable of preventing or at least reducing an increase in the sheet resistance of a specific region of a second electrode (cathode) formed after formation of the common layer structure. Therefore, the second electrode may maintain uniform voltage. That is, the first spacer structure has open regions with respect to at least some of emission parts, thus being capable of preventing an increase in the resistance of the second electrode.

[0171] Third, as the pixel structures of display devices are diversified, the first spacer structure is not limited to a uniaxial shape but has a shape surrounding the emission parts and including open regions formed in some parts of closed loop-type patterns configured to surround the emission parts so as to additionally guide the flow of current, thereby increasing the distance between the open regions of the first spacer structure between adjacent emission parts and thus being capable of improving the effect of preventing lateral leakage current.

[0172] Fourth, in addition to the first spacer structure, a second spacer structure having a positive tapered shape is provided on the bank so as to have a greater height than the first spacer structure, thereby being capable of supporting a deposition mask when the deposition mask is placed on the substrate so as to form light emitting devices, and thus protecting elements provided under the second spacer structure.

[0173] Fifth, the first spacer structure having the inverse tapered structure is disposed at least between adjacent subpixels configured to emit light with different colors and to have a threshold voltage difference therebetween, so that the common layers are disconnected at the outside of the first spacer structure, thereby preventing, when the subpixel having high threshold voltage is selectively turned on, the subpixel having low threshold voltage arranged adjacent to the turned-on subpixel from being turned on due to lateral leakage current.

[0174] It will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the spirit or scope of the invention. Thus, it is intended that the present invention cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

Claims

1. A light emitting display device comprising: a substrate comprising a first subpixel, a second subpixel, and a third subpixel arranged to neighbor one another; a first electrode at each of the first subpixel, the second subpixel, and the third subpixel; a bank that exposes the first electrode at each emission part of the first subpixel, the second subpixel, and the third subpixel; a first spacer structure on the bank; a second spacer structure spaced apart from the first spacer structure on the bank; a common layer on the first electrode, the bank, and the first spacer structure; and a second electrode on the common layer, wherein the second electrode is disposed on at least one side surface of the common layer above the first spacer structure and at least one side part of the first spacer structure, wherein the first spacer structure reduces in width in a direction toward the substrate and the second spacer structure widens in width in the direction toward the substrate, wherein each emission part of the first subpixel, the second subpixel, and the third subpixel is polygonal, wherein an upper surface of the first spacer structure has a first width and is disposed along at least one side of each emission part of the first subpixel, the second subpixel, and the third subpixel, and wherein the first spacer structure includes a first pattern surrounding the emission part of the first subpixel and having a plurality of open regions in the first pattern and a second pattern between the first subpixel and the second subpixel and, the second pattern partially parallel to the first pattern.
2. The light emitting display device according to claim 1, wherein the second spacer structure is adjacent to at least one of the plurality of open regions in the first pattern.
3. The light emitting display device according to claim 1, wherein an upper surface of the second spacer structure has a second width that is larger than the first width.
4. The light emitting display device according to claim 1, further comprising: an encapsulation layer on the second electrode, wherein a thickness of the encapsulation layer on the at least one side part of the first spacer structure is thinner than a thickness of the encapsulation layer on the upper surface of the first spacer structure.
5. The light emitting display device according to claim 1, wherein the first electrode is partly overlapped with the second spacer structure.
6. The light emitting display device according to claim 1, wherein the second pattern includes at least one open region and the second pattern partially surrounds all sides of the emission part of the second subpixel.
7. The light emitting display device according to claim 6, wherein the first spacer structure further comprises a third pattern along at least one side of the third subpixel, wherein the third pattern is parallel to the first pattern between the first subpixel and the second subpixel, and the third pattern includes at least one open region.
8. The light emitting display device according to claim 7, wherein the first pattern is disposed between the second pattern and the third pattern.
9. The light emitting display device according to claim 7, wherein the at least one open region and the second pattern and the at least one open region are positioned along a first direction.
10. The light emitting display device according to claim 7, wherein the first spacer structure further comprises, between the second subpixel and the third subpixel: a first protrusion that protrudes from the second pattern toward the third subpixel, the first protrusion being spaced apart from the third pattern by a first distance such that the first protrusion does not contact the third pattern; and a second protrusion that protrudes from the third pattern toward the second subpixel, the second protrusion being spaced apart from the second pattern by a second distance such that a third protrusion does not contact the second pattern.
11. The light emitting display device according to claim 1, wherein emission parts of the first subpixel, the second subpixel, and the third subpixel have a different shape, respectively.

- 12.** The light emitting display device according to claim 1, wherein the second spacer structure has a higher height than the first spacer structure.
- 13.** The light emitting display device according to claim 1, wherein: the first subpixel has a first emission layer that contacts the common layer between the first electrode and the second electrode so as to emit light with a first color; the second subpixel has a second emission layer that contacts the common layer between the first electrode and the second electrode so as to emit light with a second color having a shorter wavelength than a wavelength of the first color; and the third subpixel has a third emission layer that contacts the common layer between the first electrode and the second electrode so as to emit light with a third color having a longer wavelength than the first color.
- 14.** The light emitting display device according to claim 1, wherein the first width is less than a width of a smallest emission part among the first subpixel, the second subpixel, and the third subpixel.
- 15.** The light emitting display device according to claim 1, wherein the common layer comprises at least one of a hole injection layer, a hole transport layer, an electron blocking layer, an emission layer, a hole blocking layer, an electron transport layer, an electron injection layer, and a charge generation layer.
- 16.** The light emitting display device according to claim 1, wherein a threshold voltage of the second subpixel is higher than a threshold voltage of the second subpixel.
- 17.** The light emitting display device according to claim 1, wherein the first spacer structure further comprises a connection part that is between the first subpixel and another first subpixel and connects the first pattern of the first subpixel and a first pattern of another first subpixel.
- 18.** The light emitting display device according to claim 1, wherein each of the plurality of open regions in the first subpixel is shorter than a length of one side of the emission part of the first subpixel.
- 19.** The light emitting display device according to claim 1, wherein: the second subpixel and the third subpixel are alternately arranged in rows and columns; and the first subpixel is arranged to alternate with each of the second subpixel and the third subpixel in a diagonal direction.
- 20.** The light emitting display device according to claim 1, wherein the first spacer structure further comprises a fourth pattern independent from each of the first pattern and the second pattern, and wherein the fourth pattern comprises a bar-shape pattern or a loop pattern.
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